

Generation Interconnection System Impact Study Report

For Queue Project Q15-0520

Aspen 138 kV
December 24, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0520. The Interconnection Customer proposes a 84.8 MW solar photovoltaic facility interconnecting at the Aspen 138 kV substation in the Virginia area. The request was received on September 22, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 265 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,885,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Grottoes-Drain 138 kV	Dayton Substation-Crandall 345 kV Line	119.0	234	5.3
Lilly Del Caribe-Pillar Substation 138 kV	Sherburne-Bath County 345 kV	114.7	267	6.7
Shady Glen-Pillar 138 kV	Bath County-Sherburne 345 kV	114.3	265	6.7
Cairo Bend-Shady Glen Substation 138 kV line	Bath County-Sherburne 345 kV	114.3	264	6.7
Cairo Bend-Ashley 138 kV L	Bath County-Sherburne 345 kV	114.2	262	6.7
Sherburne-Ashley 138 kV	Bath County-Sherburne 345 kV L	113.8	261	6.7
West Loop-Drain 138 kV	Crandall-Dayton 345 kV Line	112.9	215	5.3
Williams Lake-Philpott 138 kV	Sherburne Substation-Pinetree 345 kV line	109.6	255	7.1
Philpott-Folsom 138 kV	Pinetree-Sherburne 345 kV	109.0	262	7.1
Dpc Prairie Du Chien-Tampa Bay 138 kV	Bath County Substation-Sherburne 345 kV	108.1	219	5.2
Mackay-Seminole Electric Cooperative 138 kV line	Sherburne-Pinetree Substation 345 kV	108.0	297	7.0
Reach-Candelaria Substation 138 kV Line	Aspen 345/138kV Transformer Bank	106.1	240	6.8
Roslyn-Tampa Bay 138 kV	Bath County-Sherburne 345 kV	105.1	270	6.5
N Riv D.P-W Loop 138 kV	Crandall-Dayton 345 kV	105.1	199	5.3
BathCounty-Folsom Prison Substation 138 kV	Bath County-Sherburne 345 kV	103.8	301	5.5
Chatom-Purdy 138 kV	Frito Lay-Dpc Prairie Du Chien 138 kV	102.7	356	8.6

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Leavitt-Chancellor Row 345 kV line	Birchwood Substation-Leavitt 345 kV	101.8	1,465	6.3
Lilly Del Caribe-Garrison 138 kV L	Sherburne-Bath County 345 kV Line	101.0	305	6.7
Fluvanna P.S-Louisa Generation Facility 345 kV Line	Leavitt-Birchwood 345 kV	100.5	3,046	9.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Drain-Grottoes 138 kV Line	Rebuild line with higher-capacity conductor	\$4,302,000	11% (\$477,000)
Pillar-Lilly Del Caribe 138 kV line	Rebuild line with higher-capacity conductor	\$12,249,000	11% (\$1,307,000)
Pillar Substation-Shady Glen 138 kV	Rebuild line with higher-capacity conductor	\$8,885,000	11% (\$948,000)
Cairo Bend-Shady Glen Substation 138 kV line	Rebuild line with higher-capacity conductor	\$3,647,000	11% (\$389,000)
Cairo Bend-Ashley 138 kV L	Rebuild line with higher-capacity conductor	\$7,422,000	11% (\$792,000)
Sherburne-Ashley 138 kV	Rebuild line with higher-capacity conductor	\$6,874,000	11% (\$734,000)
West Loop-Drain 138 kV L	Rebuild line with higher-capacity conductor	\$2,039,000	11% (\$226,000)
Philpott-Williams Lake 138 kV	Reconductor line	\$2,494,000	8% (\$205,000)
Philpott-Folsom 138 kV	Reconductor line	\$3,750,000	8% (\$308,000)
Dpc Prairie Du Chien-Tampa Bay 138 kV	Reconductor line	\$1,071,000	21% (\$226,000)
Mackay-Seminole Electric Cooperative 138 kV line	Reconductor line	\$2,862,000	8% (\$224,000)
Candelaria-Reach 138 kV	Reconductor line	\$4,959,000	6% (\$320,000)
Roslyn-Tampa Bay Substation 138 kV	Reconductor line	\$3,147,000	13% (\$409,000)
N Riv D.P-W Loop 138 kV	Reconductor line	\$1,706,000	11% (\$189,000)
Bath County-Folsom Prison 138 kV L	Reconductor line	\$5,306,000	20% (\$1,062,000)
Chatom-Purdy 138 kV	Reconductor line	\$2,823,000	17% (\$493,000)
Leavitt-Chancellor Row 345 kV line	Reconductor line	\$7,632,000	1% (\$62,000)
Lilly Del Caribe-Garrison 138 kV L	Reconductor line	\$4,075,000	11% (\$435,000)
Fluvanna P.S-Louisa Generation Facility 345 kV Line	Reconductor line	\$16,629,000	0% (\$79,000)
	Total allocated to project		\$8,885,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.